

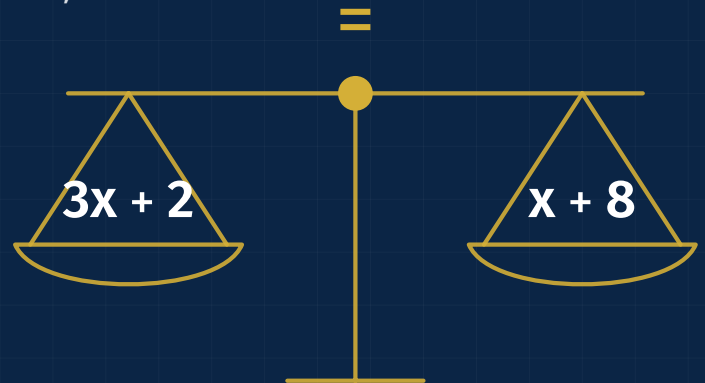


DR ESLAM AHMED
eliteigcse.com

CHAPTER 2 • ALGEBRA • TOPIC 1

Algebra Toolkit

The four tools every algebra question is built from: like terms, multiplying terms, substitution and identities.



6

VARIANTS

ALL

EVERY PAPER USES THESE

38

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Like terms

V2 Multiply & divide terms

V3 Substitution

V4 Substituting negatives

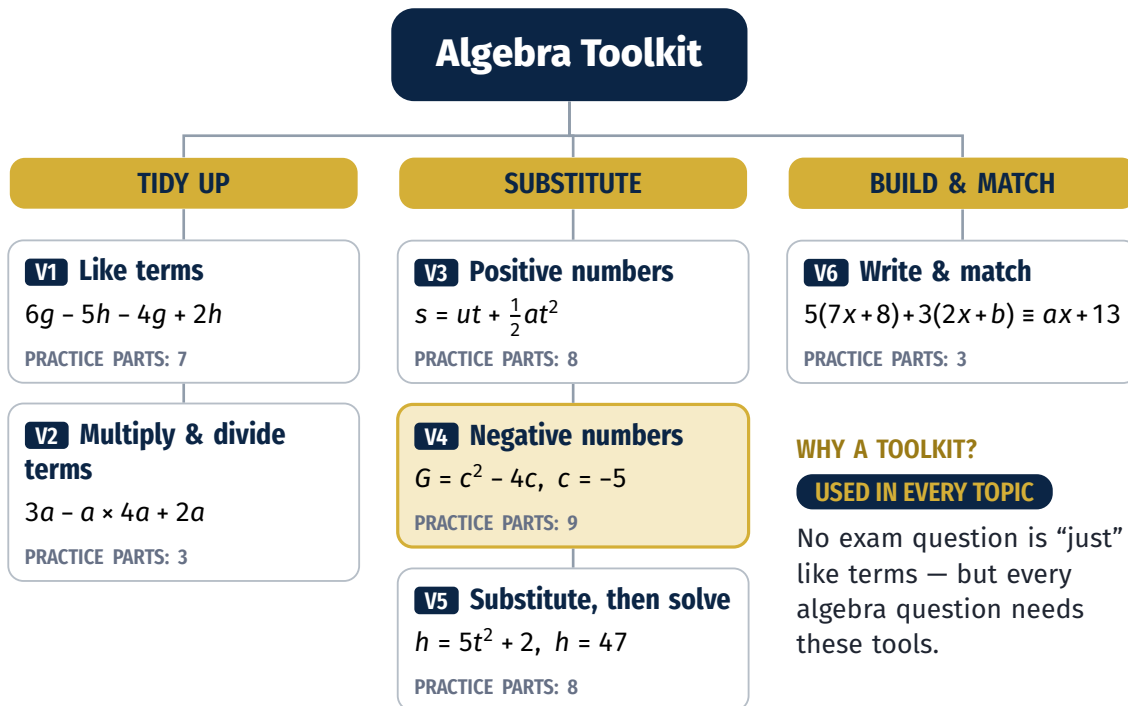
V5 Substitute, then solve

V6 Write & match expressions

TOPIC MAP · EVERY VARIANT ON ONE PAGE

Algebra Toolkit at a glance



Your notes


★ How often does each variant come up?

No past-paper question is filed under this topic: it is the toolkit inside the others. 38 practice parts, grouped by type.

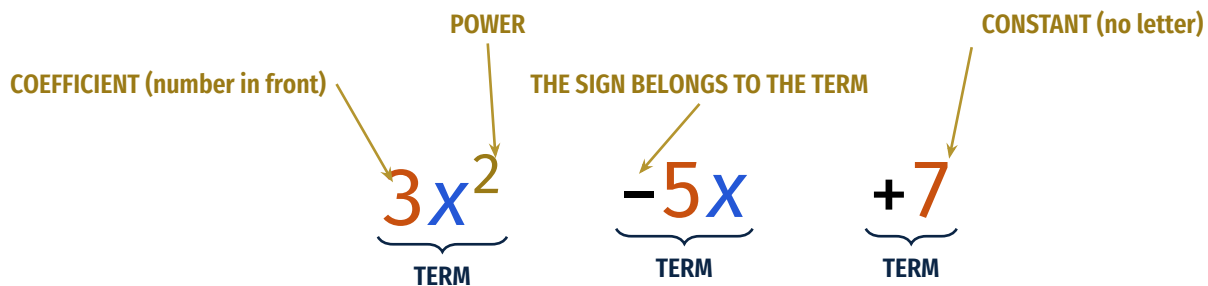
	PRACTICE PARTS
V1 Like terms	7
V2 Multiply & divide terms	3
V3 Substitution: positive	8
V4 Substitution: negatives	9
V5 Substitute, then solve	8
V6 Write & match	3

WHAT THIS TELLS YOU Substitution (V3–V5) is the biggest group, and it hides inside real exam questions — e.g. Oct/Nov 2020 Paper 1HR: “ $G = c^2 - 4c$. Find G when $c = -5$.”

BEFORE YOU START

The words of algebra



Your notes


Expression

$$3x + 5$$

Terms joined by + and -. No = sign. You **simplify** it.

Equation

$$3x + 5 = 11$$

Only some values work. You **solve** it: $x = 2$.

Formula

$$v = u + at$$

A rule linking letters. You **substitute** numbers into it.

Identity

$$2(x + 3) \equiv 2x + 6$$

True for **every** x . The sign is $\equiv$.

The order: BIDMAS

B Brackets

I Indices

D Divide

M Multiply

A Add

S Subtract

D and M have equal rank (left to right); so do A and S. $3a - a \times 4a + 2a$: multiply first.



CALCULATOR FIRST, THEN CHECK

Substitution on a Casio: type the expression with **ALPHA** **)** for x , press **CALC**, type the value (use **(** **(-)** **5** **)** for -5) and **=**. Press **CALC** again to try another value.

Check a simplification: **CALC** the question and your answer with the same value, e.g. $x = 2$. Same number? Your answer is right.

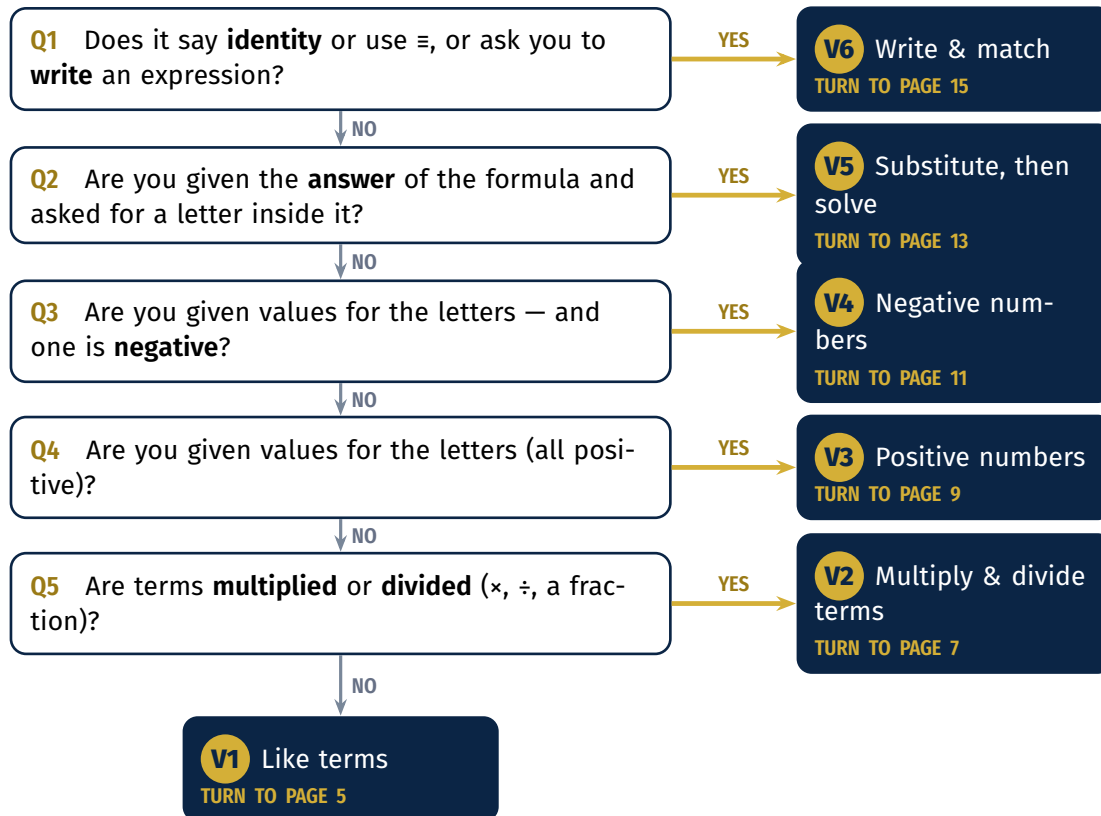
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------------------|
|  | What is it? |  | When you see $\rightarrow$ you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Collecting Like Terms

“Simplify” a long expression: put the same kinds of term together.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 7

 Your notes

? What is it?

IN PLAIN WORDS

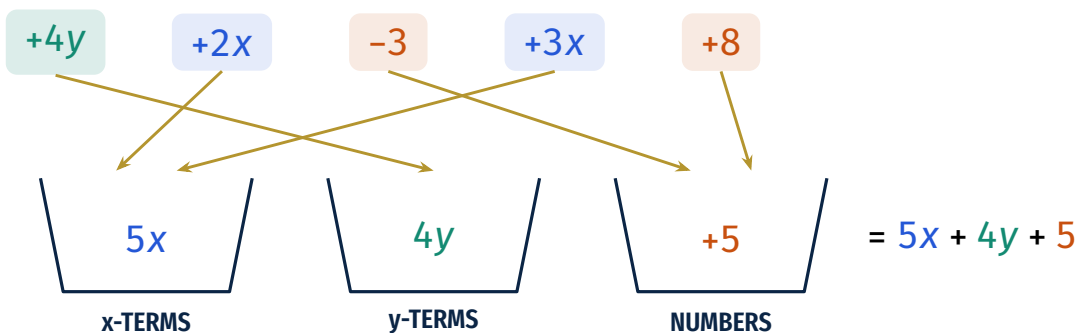
Like terms have exactly the **same letters with the same powers**: $2x$ and $3x$ are alike, x and x^2 are not. You can only add or subtract like terms — like adding apples to apples.

IT LOOKS LIKE

$$4y + 2x - 3 + 3x + 8$$

$$6g - 5h - 4g + 2h$$

$$ab - 5g + 5ab - 2g$$



How do I do it?

1 Spot like terms

Same letters, same powers. Underline each kind in its own colour — **with its sign**.

2 Group them

Write each kind together. The sign in front moves with the term.

3 Add the numbers

Add or subtract the coefficients: $2x + 3x = 5x$, $-3 + 8 = +5$.

4 Write it neatly

Letters in alphabetical order, the number last: $5x + 4y + 5$.

THE SIGN STICKS TO THE TERM -3 is “minus 3”, not 3.

V1 Collecting Like Terms

VARIANT 1 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO*"Simplify" and lots of terms*→ **Collect like terms (same letters, same powers)** *ab and ba* → **The same term: $ab + 3ba = 4ab$** *x and x^2 , x and a number*→ **Not alike — they stay separate****WORKED EXAMPLE** **PRACTICE**

Algebra Toolkit · Easy Q4 · 2 marks

Simplify $6g - 5h - 4g + 2h$ **STEP 1** $6g - 5h - 4g + 2h$

◀ spot the like terms, signs included

STEP 2 $6g - 4g - 5h + 2h$

◀ group them

STEP 3 $2g - 3h$ ◀ $6 - 4 = 2$, $-5 + 2 = -3$ **STEP 4** letters in order, nothing else to collect

◀ write it neatly

ANSWER $2g - 3h$ ✓ CALC $g = 2, h = 1$: both give 1**WATCH OUT**× $4y + 2x - 3 + 3x + 8 = 5x + 4y + 11$ ✓ $4y + 2x - 3 + 3x + 8 = 5x + 4y + 5$ The 3 has a **minus** in front: $-3 + 8 = 5$.**EXAM TIP**Check with numbers: put $x = 2, y = 3$ into the question and into your answer. Same value? Then it is right.**Check yourself**

calculator first · answers on page 17

1.1 Practice · EasySimplify
 $4y + 2x - 3 + 3x + 8$

Answer:

1.2 Practice · EasySimplify
 $ab - 5g + 5ab - 2g$

Answer:

1.3 Practice · MediumSimplify $2p - q - 3q - 5p$

Answer:

V2

VARIANT 2

Multiplying & Dividing Terms

Numbers with numbers, letters with letters
— and divide every term.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 3


Your notes

? What is it?

In plain words

Multiply: numbers $\times$ numbers, letters $\times$ letters: $a \times 4a = 4a^2$.

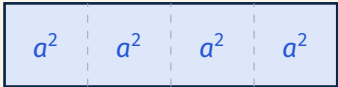
Divide a sum: the bottom divides **every term** on top: $\frac{6c + 10}{2} = 3c + 5$.

BIDMAS still rules: $\times$ and $\div$ **before** $+$ and $-$.

IT LOOKS LIKE

$$3a - a \times 4a + 2a$$

$$\frac{6c + 10}{2}$$

$$a \times 4a = 4a^2$$


$$\frac{6c + 10}{2} = \frac{6c}{2} + \frac{10}{2} = 3c + 5$$

Area model: four a -by- a squares. The 2 divides **both** terms.

How do I do it?

1 $\times$ and $\div$ first

BIDMAS: do every $\times$ and $\div$ before you add or subtract.

2 Numbers, then letters

$$3x \times 4x: 3 \times 4 = 12, x \times x = x^2 \Rightarrow 12x^2.$$

3 Divide every term

A fraction with a sum on top: divide **each** term, or factorise and cancel.

4 Collect

Now collect like terms (V1).

V2 Multiplying & Dividing Terms

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

$$3a - a \times 4a + 2a$$

→ **× first:** $a \times 4a = 4a^2$, then collect

$$\frac{6c + 10}{2}$$

→ **Divide every term:** $3c + 5$

"Show that ... always simplifies to the same value"

→ **Factorise top and bottom, cancel the bracket**

WORKED EXAMPLE PRACTICE

Algebra Toolkit · Hard Q3 · 3 marks

a and b are positive. Show that $\frac{7a + 2b - 3a}{8a + 6b + 2a - b}$ always simplifies to the same value.

STEP 1 $\frac{4a + 2b}{10a + 5b}$

◀ collect like terms, top and bottom

STEP 2 $\frac{2(2a + b)}{5(2a + b)}$

◀ factorise top and bottom

STEP 3 $\frac{2(2a + b)}{5(2a + b)} = \frac{2}{5}$

◀ the same bracket cancels

ANSWER $\frac{2}{5}$ — no a or b left

✓ try $a = 1, b = 1$: $\frac{6}{15} = \frac{2}{5}$

WATCH OUT

× $3a - a \times 4a + 2a = 2a \times 4a = 8a^2$

✓ $3a - 4a^2 + 2a = 5a - 4a^2$

BIDMAS: the $\times$ is done first; $+$ and $-$ come last.

EXAM TIP

In a "show that", your last line must be the value itself ($\frac{2}{5}$) and say it has no letters left.

Check yourself

calculator first · answers on page 17

2.1

Practice · Easy

Which is equal to
 $3a - a \times 4a + 2a$?
 $8a^2 + 2a \cdot 12a^2 \cdot$
 $5a - 4a^2 \cdot 3a - 6a^2$

Answer:

2.2

Practice · Easy

Sofia simplifies $\frac{6c + 10}{2}$:
 "divide $6c$ by 2 , then
 add 10 ". Is she right?

Answer:

2.3

Dr Eslam

Simplify $6x^2y \div 2x$

Answer:

V3

VARIANT 3

Substitution: Positive Numbers

Swap each letter for its number, then follow BIDMAS.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 8

 Your notes

? What is it?

IN PLAIN WORDS

Substitute means **replace** each letter by the number you are given. Put every number **in brackets** so nothing gets lost — then work it out with BIDMAS (or your calculator).

IT LOOKS LIKE

$$c = \frac{ma}{150}$$

$$s = ut + \frac{1}{2}at^2$$

$$B = \frac{m}{h^2}$$

$$u = 5.2$$

$$t = 7$$

$$a = 1.6$$

$$s = ut + \frac{1}{2}at^2$$



$$s = (5.2)(7) + \frac{1}{2}(1.6)(7)^2 = 36.4 + 39.2 = 75.6$$

Each letter gets its own colour and its own bracket.

How do I do it?

1 Write the formula

Copy it out first — it is your map.

2 Numbers in brackets

Replace every letter by its number, in a bracket.

3 BIDMAS

Powers first, then $\times \div$, then $+-$. Or type it all in.

4 Round at the end

Only round the final answer — to what the question asks. Add units.

NUMBERS IN BRACKETS Brackets keep every number in the right place.

V3 Substitution: Positive Numbers

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO*"Work out the value of ... when ..."*→ **Substitute, then BIDMAS***"Write down all the figures on your calculator display"*→ **Do not round — copy every digit***"Correct to 1 d.p." / "standard form, 4 s.f."*→ **Round only at the very end****WORKED EXAMPLE** **PRACTICE**

Algebra Toolkit · Medium Q13 · 2 marks

 $s = ut + \frac{1}{2}at^2$. Find the value of s when $u = 5.2$, $t = 7$ and $a = 1.6$.**STEP 1** $s = ut + \frac{1}{2}at^2$ ◀ write the formula**STEP 2** $s = (5.2)(7) + \frac{1}{2}(1.6)(7)^2$ ◀ numbers in brackets**STEP 3** $s = 36.4 + \frac{1}{2}(1.6)(49) = 36.4 + 39.2$ ◀ power first, then ×**STEP 4** $s = 75.6$ ◀ add**ANSWER** $s = 75.6$

✓ type it all in: 75.6

WATCH OUT

$$\times \frac{1}{2}at^2 = \left(\frac{1}{2} \times 1.6 \times 7\right)^2 = 31.36$$

$$\checkmark \frac{1}{2} \times 1.6 \times 7^2 = 39.2$$

The square is only on t . Square first, then multiply.**EXAM TIP**

Type the whole substitution in one go, with brackets. Then check by doing it in two parts.

Check yourself

calculator first · answers on page 17

3.1 Practice · Easy $c = \frac{ma}{150}$. Work out c when $m = 30$ and $a = 40$.

Answer:

3.2 Practice · Medium $s = ut + \frac{1}{2}at^2$. Find s when $u = 12$, $a = 10$, $t = 4$.

Answer:

3.3 Practice · Medium $x = 0.7$. Work out $\frac{(x+1)^2}{2x}$. Write all the figures on your display.

Answer:

V4

VARIANT 4

Substitution with Negatives

A negative number in a formula. Brackets save the marks.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 9


Your notes

Came up in: seen inside Oct/Nov 2020 P1HR Q3(b)

? What is it?

In plain words

Same as V3, but at least one number is **negative**. A negative must go **in a bracket**: c^2 with $c = -5$ is $(-5)^2 = 25$.

Even power → positive. **Odd power** → keeps the minus: $(-2)^3 = -8$.

IT LOOKS LIKE

$$G = c^2 - 4c$$

$$w = 5y^2 - y^3$$

$$y = mx + c$$

$$(-5)^2 - 4 \times (-5)$$

45

✓ RIGHT

$$-5^2 - 4 \times -5$$

-5

✗ WRONG

-5^2 means $-(5^2) = -25$. Only $(-5)^2$ squares the minus.

How do I do it?

1 Brackets

Every negative number goes in a bracket: (-5) .

2 Powers first

$$(-5)^2 = 25, (-2)^3 = -8.$$

3 Signs

$$- \times - = +: -4 \times (-5) = +20.$$

4 Check

Type it with brackets on the calculator — same answer?

NEGATIVES IN BRACKETS No bracket, no square of the minus.

V4 Substitution with Negatives

VARIANT 4 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO“when $c = -5$ ”→ Write (-5) every time you see c y^3 or x^3 with a negative

→ Odd power: the answer stays negative

 $y = mx + c$ with $m = -3$, $x = -2$ → $(-3)(-2) = +6$ **WORKED EXAMPLE**

PAST PAPER

Oct/Nov 2020 · Paper 1HR · Q3(b) · 2 marks

 $G = c^2 - 4c$. Find the value of G when $c = -5$.**STEP 1**

$$G = (-5)^2 - 4 \times (-5)$$

◀ every c becomes (-5) **STEP 2**

$$G = 25 - 4 \times (-5)$$

◀ power first: $(-5)^2 = 25$ **STEP 3**

$$G = 25 + 20$$

◀ $-4 \times (-5) = +20$ **STEP 4**

$$G = 45$$

◀ add; CALC agrees

ANSWER $G = 45$ ✓ CALC $c = -5$: 45**WATCH OUT** $\times -5^2 = -25$, so $G = -25 + 20 = -5$ $\checkmark (-5)^2 = 25$, so $G = 25 + 20 = 45$

Without the bracket only the 5 is squared. Bracket every negative.

**EXAM TIP**On the calculator use the $(-)$ key inside brackets: $((-) 5) x^2$.**Check yourself**

calculator first · answers on page 17

4.1

Practice · Easy

 $f = 3g + 7h$. Find f when $g = -5$ and $h = 2$.

Answer:

4.2

Practice · Medium

 $w = 5y^2 - y^3$. Find w when $y = -2$.

Answer:

4.3

Practice · Hard

 $y = x^4 - 4x^3$. Find y when $x = -1$.

Answer:

V5

VARIANT 5

Substitute, Then Solve

You know the answer of the formula — find a letter inside it.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 8

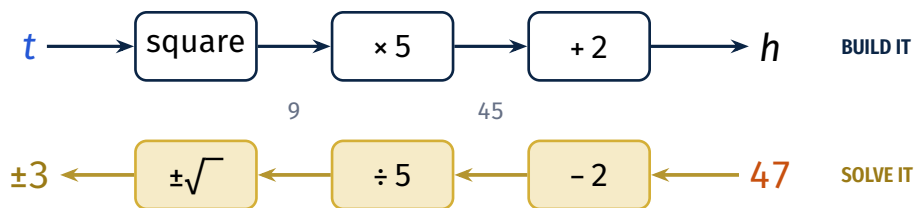
 Your notes

? What is it?**IN PLAIN WORDS**

This time you are given the **result** (the subject) and some letters, and you must find a **missing letter**. Substitute everything you know — now it is an **equation**. Solve it by undoing, last thing first.

IT LOOKS LIKE

$$A = 4bc \quad h = 3t^2 \quad h = 5t^2 + 2 \quad ax^2 + a = 150$$



Build the formula forwards; solve it backwards. A square root gives two answers.

How do I do it?**1 Substitute**

Put in every number you know, in brackets.

2 Simplify

Work out what you can. Now it is an equation.

3 Solve

Undo, last thing first: -2 , then $\div 5$, then $\sqrt{}$.

4 Square root $\rightarrow \pm$

$t^2 = 9$ gives $t = 3$ **or** $t = -3$. Check both.

SUB $\rightarrow$ SOLVE Rearranging first also works — that is Topic 7.

V5 Substitute, Then Solve

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

“Work out the value of t when $h = 108$ ” → **Substitute h , then solve for t**

“Work out **a** value of t ” → **There may be two ($\pm$) — one is enough**

“One solution of ... is $x = 7$ ” → **Put $x = 7$ in to find the unknown constant**

 WORKED EXAMPLE **PRACTICE**

Algebra Toolkit · Medium Q6(ii) · 2 marks

$h = 5t^2 + 2$. Work out a value of t when $h = 47$.

STEP 1 $47 = 5t^2 + 2$ ◀ substitute $h = 47$

STEP 2 $45 = 5t^2$ ◀ take 2 off both sides

STEP 3 $t^2 = 9$ ◀ divide by 5

STEP 4 $t = 3$ or $t = -3$ ◀ square root: $\pm$

ANSWER $t = 3$ (or -3)

✓ CALC $t = 3: 5 \times 9 + 2 = 47$

 WATCH OUT

$$\times 5t^2 = 45 \Rightarrow t = \frac{45}{10}$$

$$\checkmark 5t^2 = 45 \Rightarrow t^2 = 9 \Rightarrow t = \pm 3$$

t^2 means $t \times t$, not $2 \times t$. Divide by 5, then square root.

 EXAM TIP

If t is a time or a length, only the positive answer makes sense — say so.

 Check yourself

calculator first · answers on page 17

5.1 Practice · Medium

$A = 4bc$. $A = 100$, $b = 2$.
Work out c .

Answer:

5.2 Practice · Medium

$h = 3t^2$. Work out t
when $h = 108$ ($t > 0$).

Answer:

5.3 Practice · Hard

One solution of
 $ax^2 + a = 150$ is $x = 7$.
(a) Find a . (b) Find the
other solution.

Answer:

V6

VARIANT 6

Writing & Matching Expressions

Turn words into algebra; make two sides match for every x .

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 3

 Your notes

? What is it?

In plain words

Write: turn a sentence into an expression — “four times as many as $(x + 3)$ ” is $4(x + 3)$.

Match: an **identity** ($\equiv$) is true for **every** x , so the x -terms match and the numbers match.

IT LOOKS LIKE

$$4(x + 3) \equiv ax + 13$$

$$41x + (40 + 3b) \equiv ax + 13$$

x-TERMS MATCH
NUMBERS MATCH

$$a = 41, \quad 40 + 3b = 13 \Rightarrow b = -9$$

How do I do it?

1 Expand + simplify

Multiply out the side with brackets and collect like terms.

2 Match x -terms

The coefficients of x are equal on both sides.

3 Match numbers

The constant terms are equal on both sides.

4 Solve

Solve each small equation for the unknowns.

MATCH THE COEFFICIENTS $\equiv$ means “the same expression in disguise”.

V6 Writing & Matching Expressions

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO"identity", the sign $\equiv$ → **Expand, then match x -terms and numbers**

"Circle the expression for ..."

→ **Build it step by step in words, then algebra**"If x doubles, what happens to y ?"→ **Try numbers: $x = 1$, then $x = 2$** **WORKED EXAMPLE****PRACTICE**

Algebra Toolkit · Hard Q4 · 4 marks

Work out the values of a and b in the identity $5(7x+8)+3(2x+b) \equiv ax+13$

STEP 1 $35x + 40 + 6x + 3b \equiv ax + 13$

◀ expand both brackets

STEP 1 $41x + (40 + 3b) \equiv ax + 13$

◀ collect like terms

STEP 2+3 $a = 41, \quad 40 + 3b = 13$

◀ match the x -terms, then the numbers

STEP 4 $3b = -27 \Rightarrow b = -9$

◀ solve

ANSWER $a = 41, b = -9$

✓ $x = 1$: both sides give 54**WATCH OUT**

✗ $3(2x + b) = 6x + b$

✓ $3(2x + b) = 6x + 3b$

The 3 multiplies the b too. Then

$40 + 3b = 13$, not $40 + b = 13$.

**EXAM TIP**Check an identity with any number: $x = 1$ in both sides must give the same value.**Check yourself**

calculator first · answers on page 17

6.1

Practice · Hard

Amy has x beads, Billy 3 more than Amy, Carly 4 times Billy's. How many has Carly?

Answer:

6.2

Practice · Easy

 $y = \frac{10}{x}$. If x doubles, what happens to y ?

Answer:

6.3

Dr Eslam

Find a and b :

$3(2x + a) + bx \equiv 10x + 12$

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

1.1 $5x + 4y + 5$

1.2 $6ab - 7g$

1.3 $-3p - 4q$

2.1 $5a - 4a^2$

2.2 Wrong: 10 must be divided by 2 too, $3c + 5$

2.3 $3xy$

3.1 8 ml

3.2 $s = 128$

3.3 2.064285714

4.1 $f = -1$

4.2 $w = 28$

4.3 $y = 5$

5.1 $c = 12.5$

5.2 $t = 6$

5.3 (a) $a = 3$ (b) $x = -7$

6.1 $4(x + 3)$

6.2 y is divided by 2 ($\div 2$)

6.3 $a = 4, b = 4$

THE WHOLE TOPIC ON ONE PAGE

Algebra Toolkit cheat sheet



Your notes
V1 Like terms

$$6g - 5h - 4g + 2h = 2g - 3h$$

Same letters, same powers. The sign sticks to the term.

V2 Multiply & divide

$$a \times 4a = 4a^2$$

$\times \div$ before $+-$. Divide **every** term on top.

V3 Substitution

$$s = (5.2)(7) + \frac{1}{2}(1.6)(7)^2$$

Numbers in brackets, BIDMAS, round at the end.

V4 Negatives

$$(-5)^2 = 25, \quad -5^2 = -25$$

Bracket every negative. Even power $\rightarrow +$, odd power keeps $-$.

V5 Substitute, then solve

$$47 = 5t^2 + 2 \Rightarrow t = \pm 3$$

Put the numbers in, then undo last-thing-first. $\sqrt{\quad}$ gives $\pm$.

V6 Write & match

$$41x + (40 + 3b) \equiv ax + 13$$

$\equiv$: match x-terms and numbers.

**TOP 3 LOST MARKS**

1. Losing a minus: $-3 + 8 = 11 \times$
2. No bracket on a negative: $-5^2 \times$
3. Adding before multiplying:
 $3a - a \times 4a = 2a \times 4a \times$

**CALCULATOR FIRST, THEN CHECK**

Type the expression with **ALPHA** **)**, press **CALC**, enter the value.
 Check a simplification: same value before and after.

BOTTOM LINE Like terms · BIDMAS · brackets round every number you substitute · $\equiv$ means match.